HDOC DAY 63

PROBLEM

Write a C program that reads the size and elements of a variable-length integer array, nums[]. The array contains distinct integers from 0 to size, with exactly one number missing. Pass the array to a function named missing_number() using its address. Use pointers to pass the array between the calling function and missing_number(). The function missing_number() must:

- Find the missing integer from the range 0 to size.
- Display the missing number.
- Assume that all elements in the array are distinct.
- Use the fact that the array contains exactly n numbers from the range 0 to size, with one number missing.
- Display the result as an integer.

CODE

```
#include <stdio.h>


int missing_number(int *nums, int size)
{
   int expected_sum = size * (size + 1) / 2;
   int actual_sum = 0;


   for (int i = 0; i < size; i++)
   {
      actual_sum += *(nums + i);
   }


   return expected_sum - actual_sum;
```

```
}

int main()
{
   int size;

   printf("Enter size: ");
   scanf("%d", &size);

   int nums[size];

   printf("Enter %d elements: ", size);

   for (int i = 0; i < size; i++)
   {
      scanf("%d", &nums[i]);
   }

   printf("%d\n", missing_number(nums, size));

   return 0;
}
```